Statement of Purpose
Amazon ML Summer School 2026 | Aayush

I had the neural network I built without PyTorch or TensorFlow and just using NumPy, finally starting to learn, when it was past midnight. I learned a lot from that loss curve dropping off nicely, not only that it worked but how, the mathematics of backpropagation, the geometry of gradient descent. Thereafter, I began to regard ML as a discipline and not as a toolbox.

I'm a third year B.Tech Computer Science student at I K Gujral Punjab Technical University (CGPA-9.43) and am expected to graduate in 2027. My background is in probability, statistics, linear algebra, and optimization, which are the mathematical pillars of MLSS. The results are published on Hugging Face Hub, with a fine-tuned version of the DistilBERT model that achieves a 91% F1-score, a 15% improvement over traditional baselines, and an Enterprise RAG AI Assistant that processes 1,000+ PDF pages across 50,000+ embeddings, reducing retrieval latency from 4.8s to under 1.5s, with a 35% improvement in multi-turn query accuracy, applied to real systems.

Where my focus lies is in the context of language acquisition; from intent to understanding, turn-to-turn shifts in meaning, and where current architectures don't quite get it. This was revealed by building the RAG assistant, where the retriever was not incorrect, it was merely that the model was not grounded enough to understand exactly what the user was looking for throughout the turns. I'm attracted to NLP because of that unsolved problem, the probabilistic modelling, and because of the connection between learned representation and real intent.

That's what I want to fill with Amazon ML Summer School. The course material is relevant: supervised learning, deep learning, probabilistic graphical models, and NLP are all areas where I feel I have gaps in theoretical understanding that I can strengthen through the program. More importantly, learning from Amazon Scientists who build production-scale systems will expose me to challenges that benchmarks rarely capture: distribution shift, latency-accuracy trade-offs, and failure modes at scale.

I do not wish to add a credential. I'm applying because the problems that I'm facing — why attention degrades on long-range dependencies, how probabilistic models deal with real-world noise, and what it takes to align model outputs with true user intent — are precisely the problems this program is built to address. I'm eager to learn, contribute, and grow through this opportunity.